

Sateesh Kumar Nunna

+1-405-413-4413 | Sateesh195.n@gmail.com | [LinkedIn](#)

OBJECTIVE

Senior Data Engineer with extensive experience designing and implementing cloud-based data pipelines using Python, SQL, Snowflake and Spark. Developed scalable ETL and real-time streaming workflows on AWS, GCP and Azure to integrate diverse sources and support analytics and machine learning use cases. Collaborated closely with analysts, marketers and IT teams to deliver reliable, timely data solutions that drive business insights.

Technical Skills

- **Big Data Technologies:** HDFS, Apache Spark, Hadoop, Hive, Pig, Sqoop, Impala, HBase, Kafka, Flume, Yarn, Cloudera Manager
- **Programming Languages:** Python, Java, Scala, SQL, JSON, HTML
- **Cloud Platforms:** AWS (EC2, S3, Glue, Redshift, Athena, Lambda, EMR, Kinesis), Azure (Data Factory, Databricks, Azure SQL, Synapse Analytics, Azure Data Lake), GCP (BigQuery, GCS, Dataflow, Cloud Functions, Pub/Sub), AWS App Flow, AWS DMS, Google Cloud Platform
- **Data Warehousing:** Snowflake, Redshift, Teradata, Oracle, MySQL, PostgreSQL, SQL Server, MongoDB, Data Modeling, Dimensional Design, Data Lakehouse Architectures
- **Data Engineering Tools:** Apache Airflow, Talend, SSIS, Azure Data Factory, ETL Pipelines, Control-M, Data pipelines, CRM Systems Integration
- **DevOps CI/CD:** Docker, Kubernetes, Terraform, Jenkins, Git, GitHub, Bitbucket, CloudFormation, Infrastructure As Code
- **Data Processing Streaming:** Spark Streaming, Structured Streaming, Apache Beam, Kafka, ELK Stack (Elasticsearch, Logstash, Kibana)
- **Machine Learning Analytics:** Databricks MLflow, TensorFlow, PyTorch, Scikit-learn, Power BI, Tableau, Matplotlib, ggplot2, Looker, Google Analytics 4
- **Security Compliance:** Azure Active Directory (AAD), Role-Based Access Control (RBAC), Multi-Factor Authentication (MFA), FERPA Compliance
- **Collaboration Agile Tools:** JIRA, Confluence, Slack, Microsoft Teams, Agile (Scrum, Kanban, SAFe)
- **Scripting Automation:** PowerShell, Shell Scripting, Python Scripting, Boto3 for AWS Automation
- **Web Application Servers:** Apache Tomcat, WebLogic, WebSphere
- **Monitoring Logging:** Nagios, CloudWatch, Splunk, Prometheus, Grafana
- **Development Environments:** Eclipse, Visual Studio, IntelliJ IDEA, PyCharm, Jupyter Notebook
- **Data Science:** Data science
- **Data Formats:** Parquet, Avro

Experience

Bank of America (Infosys)

Aug 2025 - Present

AI Data Engineer

Plano, USA

- Designed and developed AI-driven data pipelines using Python and PySpark, implementing robust data modeling to enable large-scale processing of financial datasets for machine learning applications
- Built and optimized data ingestion and transformation workflows using Sqoop, Apache Spark, and Tableau, reducing data latency and supporting AI/ML model training and real-time analytics
- Developed feature engineering pipelines using PySpark, improving model accuracy and performance for financial risk and loan processing use cases
- Integrated machine learning workflows with data pipelines, enabling seamless deployment of predictive models into production environments
- Processed and transformed large-scale datasets with Hadoop HDFS and Hive, creating ready-to-train data that cut preprocessing time for distributed AI model training
- Implemented Python-based data validation and preprocessing pipelines using Pandas and Great Expectations, reducing data errors by 30% and delivering clean input for AI/ML models
- Built real-time data processing pipelines with Kafka and Spark Streaming, achieving sub-second latency for event-driven AI applications and enabling timely insights
- Collaborated with Data Scientists and ML Engineers to deploy and fine-tune AI models using Docker and CI/CD pipelines, increasing model inference speed by 25% and improving decision-making
- Developed AI-powered data solutions with scikit-learn, TensorFlow, and MLflow for model tracking, delivering predictive analytics that automated 15% of routine reporting tasks

Continental Resources

Nov 2023 - Aug 2025

Azure Data Engineer

Oklahoma, USA

- Engineered and optimized ETL workflows using Azure Data Factory and Talend to integrate data across Azure Data Lake, Blob Storage, and SQL Database, enhancing accessibility and reliability.

- Built scalable data pipelines with Azure Databricks, PySpark, and Spark Streaming for real-time data processing and analytics, supporting faster insights and decision-making.
- Implemented data ingestion frameworks using Kafka, Snowflake, and API integrations to handle both structured and semi-structured data, incorporating modern data lakehouse architectures and improving accuracy and processing speed.
- Automated cloud resource provisioning using Terraform, reducing manual errors and deployment times through effective Infrastructure as Code practices.
- Enhanced query performance on Azure Synapse Analytics and Snowflake by applying indexing, partitioning, and clustering strategies, resulting in noticeably faster query execution and improved data retrieval speed.
- Deployed CI/CD pipelines using Jenkins, Docker, and Kubernetes, which reduced deployment time and increased system reliability.
- Developed real-time streaming analytics solutions using Azure Stream Analytics, Kafka, and Spark Streaming to drive quick decision-making.
- Utilized Python and SQL in Databricks and Azure Synapse Analytics to transform, validate, and process data, improving data quality and reducing processing errors.
- Managed Snowflake data warehouse by applying data governance and security controls, maintaining data integrity and meeting compliance standards.
- Conducted data migration between on-premises and cloud environments using Azure Data Factory, SSIS, and Python scripts, which minimized downtime and prevented data loss.
- Enhanced business insights by developing interactive Tableau dashboards for improved data visualization and reporting.
- Strengthened data security and compliance through the implementation of Azure Active Directory, Role-Based Access Control, and Multi-Factor Authentication.
- Scheduled and optimized workflow automation using Control-M and Airflow, boosting efficiency in processing data pipelines.
- Improved system troubleshooting and reliability by implementing cloud monitoring and logging with Azure Monitor, Log Analytics, and the ELK Stack.

BOK Financial Corporation

AWS Data Engineer

Dec 2022 - Oct 2023

Tulsa, Oklahoma, USA

- Developed and optimized AWS Glue ETL pipelines and AWS App Flow that moved data between Amazon S3, Redshift, and Snowflake, improving data availability for downstream analytics.
- Designed and deployed Python-based AWS Lambda functions to validate, cleanse, and transform streaming data, automating tasks and reducing manual effort.
- Built and maintained Apache Spark applications with Scala and PySpark on EMR to process financial datasets, accelerating data processing workflows.
- Engineered a Kafka event-driven architecture on AWS MSK to stream transactional data in real time, enabling timely insights for business users.
- Orchestrated complex ETL workflows in SSIS using conditional branching, error handling, and logging, which increased workflow reliability and simplified troubleshooting.
- Enhanced query performance and storage efficiency through the development of data partitioning strategies in Hive and Redshift.
- Secured data access and ensured regulatory compliance by implementing AWS IAM policies and role-based access control (RBAC).
- Automated AWS infrastructure provisioning using Terraform, increasing deployment efficiency and scalability.
- Led data migration initiatives from Amazon Redshift to Snowflake to optimize query execution and improve warehouse performance.
- Deployed containerized data processing solutions by designing and managing Kubernetes clusters for scalability and high availability.
- Configured and monitored Control-M workflows, ensuring timely execution of critical data jobs with effective failure notifications.
- Developed Spark Streaming applications for processing Kafka messages, enabling near real-time data analytics and reporting.

Lloyds Banking Group

Sr. GCP Data Engineer

Jun 2021 - Jul 2022

London, UK

- Designed and implemented data pipelines with Google Cloud Dataflow and Apache Beam, integrating Google Analytics 4 data to enable faster ingestion and transformation of large datasets.
- Optimized query performance and reduced latency for analytics workloads by migrating Oracle databases to Google BigQuery.
- Automated ETL pipeline orchestration and monitoring by building GCP Cloud Composer (Airflow) workflows, which reduced manual intervention and improved reliability.
- Implemented cost optimization strategies by analyzing Google Cloud billing reports and tuning BigQuery queries.
- Built real-time data streaming solutions with Pub/Sub, Dataflow, and Apache Kafka to accelerate decision-making for financial risk assessments.
- Managed HBase NoSQL databases integrated with PySpark for improved data retrieval and real-time processing efficiency.
- Streamlined data validation and transformation tasks by developing serverless applications using Google Cloud Functions.
- Provided actionable insights to stakeholders by creating and optimizing Power BI dashboards for real-time operational views.
- Enhanced cloud security and compliance using Google VPC and IAM policies to enforce data protection protocols.
- Improved data reliability by conducting quality assessments and implementing automated validation checks across data lakes and warehouses.
- Collaborated with cross-functional teams to modernize legacy ETL workflows, reducing processing times by 30%.

- Supported fraud detection and risk analysis by integrating Scikit-learn machine-learning models into BigQuery ML, enabling faster risk scoring and improving detection accuracy

Teleperformance

Oct 2017 - Jul 2019

Data Engineer

Hyderabad, India

- Designed and optimized ETL/ELT pipelines to integrate, process, and transform high-volume data from multiple sources into Azure Data Lake, Blob Storage, and SQL Data Warehouse, which reduced data load time and improved query performance
- Developed custom logging frameworks for monitoring ELT pipelines in Azure Data Factory, improving data visibility and troubleshooting efficiency.
- Implemented real-time data streaming solutions using Kafka and Spark Streaming to ensure seamless data ingestion for business intelligence and analytics, cutting data latency and enabling more timely reporting
- Created and deployed Databricks Spark jobs using PySpark for large-scale data transformation and machine learning model training, which accelerated model training cycles and delivered faster analytical insights
- Engineered SQL Server Reporting Services (SSRS) reports with drill-down, parameterized, and matrix capabilities to provide actionable insights.
- Automated data migration from on-premises databases to Azure Data Lake Store using Azure Data Factory, enhancing enterprise data scalability and accessibility.
- Optimized resource utilization in cloud-based data processing workflows by designing performance-tuning strategies for Kubernetes and Docker deployments, which reduced processing time and lowered cloud costs
- Integrated Apache Airflow for job scheduling and built DAGs to orchestrate and monitor end-to-end data pipelines using Python, improving pipeline reliability and decreasing job failures
- Ensured data consistency and accuracy by conducting comprehensive quality checks across Teradata, Oracle, and Azure SQL environments, resulting in fewer data errors and higher trust in reports
- Implemented CI/CD pipelines with Jenkins and Git to streamline ETL deployment processes, enhancing development and release cycles.
- Managed Azure Databricks notebooks to automate ETL transformations and boost data processing efficiency.
- Collaborated with cross-functional teams using Looker and Git to implement data governance, compliance, and security protocols for sensitive data assets, ensuring regulatory compliance and reducing data access errors

Education

Oklahoma Christian University

Aug 2022 - Sep 2023

Master's, computer science

Certifications

- **Databricks Certified: Generative AI Engineer Associate**
- **Databricks Certified: Data Engineer Associate**
- **Microsoft Certified: Azure Data Fundamentals.**
- **Google Cloud Certified: Associate Cloud Engineer.**
- **AWS Certified: Data Engineer - Associate.**